A Neuro-Symbolic Expert System that Learns the Rules: Fusing a Differentiable Rule Layer with Neural Tabular Models for Interpretable, Calibrated, and Robust Decision Support

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Abstract

Rule-based expert systems are interpretable, auditable, and data-efficient, but they cannot learn from data and degrade when the encoded knowledge is approximate. Deep tabular models are accurate and flexible, but they are opaque, hard to calibrate, and brittle under distribution shift. We study how far the two paradigms can be reconciled in a single, deployable model and introduce NEXS (*Neuro-symbolic EXpert System*), which couples a differentiable rule layer—whose per-rule confidences are initialised and regularised towards an expert's certainty factors—with a neural tabular head, fused through an additively-gated logit. Unlike semantic-loss approaches that use knowledge only as a soft consistency penalty on the output, NEXS keeps the symbolic computation in the forward pass, so the learned logic remains explicit and recoverable. We build two controlled benchmarks—clinical triage and industrial fault diagnosis—in which the data are generated by a *known* Horn rule base whose intermediate concepts are hidden, enabling the simultaneous measurement of task accuracy and *rule recovery*. Across three seeds, NEXS recovers the most ground-truth rules of any learned model (rule-set F1 up to 0.33 versus 0.11–0.22 for an MLP and semantic-loss model and 0 for logistic regression) and attains the highest logic fidelity to the ground-truth oracle among learned models (0.86–0.91, approaching the expert system's 0.95–0.97). It matches the MLP on accuracy and is the best learned model in low-data regimes—up to +7 F1 points over the MLP at 300 training samples— while running at $\sim$0.6 μs per sample, about 110$\times$ faster than a random forest. We also find that a classical certainty-factor expert system, despite being the most logic-faithful and shift-robust predictor, is the *worst* calibrated (ECE $\sim$0.09), because its certainty factors are heuristics rather than probabilities; isotonic calibration makes all learned models well calibrated. Finally, we repeat the study on three small public datasets—UCI Cleveland heart disease, Pima diabetes, and tic-tac-toe—with hand-written knowledge bases. There the expert system is exact only when its knowledge is exact (tic-tac-toe, where it is the eight winning lines) and is the *weakest* method when the knowledge is approximate, while NEXS retains the most expert logic of any learned model. We release code, data, and a fully reproducible experiment suite.

1 Introduction

Expert systems encode human knowledge as explicit rules and dominated applied AI in the 1970s–80s, from MYCIN's certainty-factor calculus [20] to large diagnostic shells. Their strengths are well

understood: decisions are traceable, they work with little or no data, and they degrade gracefully when rules are approximate. Their weaknesses are equally well understood: they cannot learn from data, knowledge acquisition is a bottleneck, and a wrong rule is a permanent error. The deep-learning era inverted the trade-off. Gradient-boosted trees and multilayer perceptrons now dominate tabular decision tasks, but they are opaque, miscalibrated [9], and brittle to distribution shift.

The third wave of neuro-symbolic AI [8, 12] argues that these paradigms should be combined rather than opposed. A large body of work does so by *compiling* knowledge into a differentiable loss: semantic loss [22], logic-rule distillation [10], and DL2 [5] all penalise outputs that violate the knowledge base during training. Others embed symbolic reasoning directly in the computation graph, as in DeepProbLog [15] and logic tensor networks [1], or inject rules into the architecture, as in the KBANN lineage [21, 6]. Yet a practical question remains open: *for the tabular decision-support problems that industry actually deploys, which coupling is worth its complexity, and does the resulting model still behave like the expert system it replaces?*

Answering this requires measuring more than accuracy. A model that replaces an expert system must (i) be at least as accurate, (ii) be calibrated enough to threshold safely, (iii) remain robust when the input distribution moves, and (iv) *learn logic that an expert can inspect and trust.* Most neuro-symbolic evaluations report (i) and sometimes (ii) or (iii); almost none report (iv), because doing so needs ground-truth rules, which real datasets rarely provide.

Contributions.

- We design two reproducible benchmarks—*clinical triage* and *industrial fault diagnosis*—in which data are generated by a known Horn rule base with *hidden* intermediate concepts, giving a ground-truth oracle for both accuracy and logic (Section 3).
- We introduce NEXS, a neuro-symbolic expert system that keeps a differentiable rule layer in the forward pass and anchors each rule's confidence to an expert certainty factor via a prior regulariser (Section 4.4).
- We run a seven-experiment study (Section 5) comparing a certainty-factor expert system, four classical learners, an MLP, a semantic-loss model, and NEXS, measuring accuracy, calibration, rule recovery, logic fidelity, data efficiency, shift robustness, and latency.
- We show that NEXS recovers more ground-truth rules and is more logic-faithful than any other learned model, matches the MLP on accuracy, and is strongest in low-data regimes, while the discrete expert system is the most logic-faithful and shift-robust but the *worst* calibrated predictor (Section 6).

2 Related Work

Neural-symbolic integration. The field is surveyed in [2, 7, 13, 16]. Approaches differ in *where* knowledge enters. Loss-based methods add a knowledge penalty to the objective: semantic loss [22] measures the probability mass of worlds satisfying a constraint; rule distillation [10] projects predictions onto a rule-regularised space; DL2 [5] trains with logical constraints as inequalities. Architecture-based methods build rules into the network: KBANN [21] initialises a network from propositional rules, and logic tensor networks [1] ground first-order logic in a real-valued tensor calculus. Probabilistic methods such as DeepProbLog [15] couple neural predicates with a logic program and perform inference in a probabilistic logic. NEXS sits between the loss-based and

architecture-based families: like KBANN it uses the knowledge base to structure and initialise the computation, but it does so through a *differentiable, continuously-relaxed* rule layer whose confidences stay tied to the expert's certainty factors.

Rule extraction and interpretability. Post-hoc explainers such as LIME [18] and SHAP approximate a black box locally; recent work extends these to multivariate time series [14]. Extracting rules by distillation, as we do to evaluate logic recovery, is an established evaluation device but is rarely compared against ground-truth rules. Our rule-recovery metric closes that gap.

Hybrid AI and expert systems. Surveys of hybrid AI catalogue how rules and models are coupled in deployed systems [19], and applied reviews emphasise the need for interpretability in high-stakes domains [4]. We contribute a quantitative, multi-criteria comparison on controlled data.

3 Problem Formulation and Benchmarks

We study binary decision support where the label is the disjunction of a small number of conjunctive rules over *observed features* and *latent intermediate concepts*. Formally, a domain is a tuple $\langle \mathbf{f}, \mathbf{c}, \mathcal{R} \rangle$ where $\mathbf{f}$ are observed features, $\mathbf{c}$ are hidden concepts, and $\mathcal{R}$ is a Horn rule base whose heads are concepts or the label. Data are generated by first sampling features from a two-regime mixture (a shared latent propensity drives a "healthy" and an "at-risk" regime per feature), then realising concepts from the concept rules, then the label from the label rules, with small stochastic realisation and label noise. Concepts are never revealed to any model.

Domains. *Clinical triage* (8 features: age, systolic BP, cholesterol, max heart rate, ST depression, BMI, smoking, diabetes; concepts: hypertension, dyslipidemia) predicts high cardiac risk, in the spirit of the UCI heart-disease task [11]. *Industrial fault diagnosis* (6 sensors: temperature, pressure, vibration, flow, current, RPM; concepts: overheat, bearing wear) predicts an equipment fault, following predictive-maintenance benchmarks [17]. Each domain has five ground-truth rules, three of which produce the label. An *expert* rule base with human-plausible (rounded) thresholds and certainty factors is given to the symbolic and neuro-symbolic methods; it deliberately differs from the ground truth, so we can measure tolerance to imperfect knowledge.

4 Methods

4.1 Symbolic expert system (ESS)

We implement forward chaining in two layers. Concept confidences are computed from concept rules, then the label from concept and feature rules. A literal's membership is crisp (a step function) or soft (a tanh ramp), and contributions to the same head are combined with the MYCIN certainty-factor calculus [20]: $\mathrm{cf}(h) \leftarrow a + b(1-a)$ for two contributions a, b. This is the classical, fully transparent baseline.

4.2 Classical and neural baselines

We compare logistic regression, a depth-limited decision tree, a random forest, and gradient boosting [3], plus a multilayer perceptron (MLP) with two hidden layers. All models are trained on the

same splits; probabilistic outputs are isotonic-calibrated on a validation split and thresholded at the F1-optimal validation threshold.

4.3 Semantic-loss model

Following [22], an MLP head is trained with the task loss plus a knowledge penalty. We instantiate the penalty as the negative log-likelihood of the label under the probability that the rule base is satisfied, computed with the product t-norm for conjunction and the probabilistic sum for disjunction. Here the knowledge shapes the *loss* only; the decision function remains the MLP.

4.4 Proposed: NEXS

NEXS keeps the symbolic computation in the forward pass. A differentiable rule layer maps inputs to a label probability p_{sat} using the same t-norm relaxation, but each rule r carries a learnable confidence c_r, initialised to the expert's certainty factor and *regularised towards it*. A neural tabular head produces a logit $f(\mathbf{x})$, and the two are fused additively:

$$z(\mathbf{x}) = f(\mathbf{x}) + \alpha \cdot \operatorname{logit}\big(p_{\text{sat}}(\mathbf{x})\big), \qquad \alpha \geq 0. \tag{1}$$

The training objective is

$$\mathcal{L} = \operatorname{BCE}\big(\sigma(z), y\big) + \lambda_{\text{sem}} \operatorname{BCE}\big(p_{\text{sat}}, y\big) + \lambda_{\text{prior}} \tfrac{1}{|\mathcal{R}|} \textstyle\sum_r (c_r - c_r^\star)^2, \tag{2}$$

where $c_r^\star$ is the expert certainty factor. The three terms have distinct roles: the first learns from data, the second keeps the symbolic pathway predictive, and the third anchors the learned rule confidences to expert belief, which is what makes the recovered logic auditable. Setting α=0 removes the rule layer, λ_{sem}=0 removes the semantic term, and λ_{prior}=0 removes the expert anchor; we use these as ablations, together with a *rule-layer-only* variant ($f \equiv 0$) that tests the symbolic pathway in isolation.

5 Experimental Setup

For each domain we draw 4000 samples and split them 60/20/20 into train/validation/test, repeated over three seeds. We report the following.

E1 - Task performance accuracy, minority-class F1, ROC-AUC, and expected calibration error (ECE, 10 bins).

E2 - Rule recovery we distil a depth-limited decision tree from each model on uniform grid samples and match the extracted label rules against the ground-truth logic. Because the true label rules reference hidden concepts, the reference is the *concept-expanded feature-only DNF* obtained by substituting each concept with its rule bodies and distributing conjunctions; a literal matches when it refers to the same variable and defines the same partition of the feature domain (up to 6% of a fine grid), which makes the comparison robust to operator-equivalent forms such as $x \geq 0.5$ and $x > 0.45$. We report rule-set F1.

E3 - Logic fidelity agreement between a model's decisions and the deterministic oracle built from the ground-truth rules, measured on a uniform grid that probes regions away from the training support.

E4 - Data efficiency minority-class F1 as training size grows from 150 to 2400.

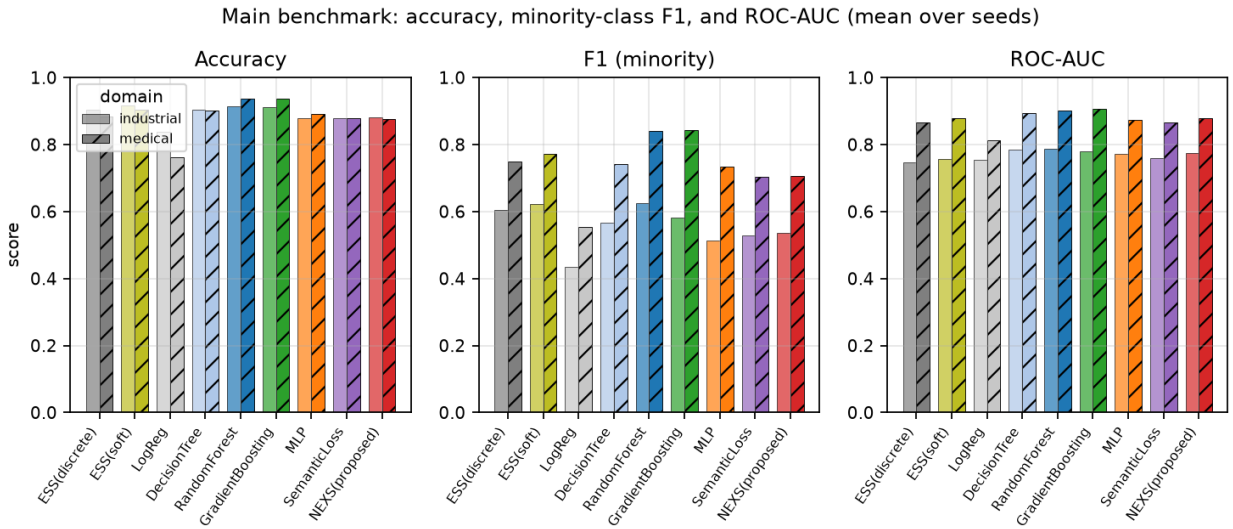


Figure 1: Accuracy, minority-class F1, and ROC-AUC by method and domain.

E5 - Robustness accuracy under a covariate shift and under a novel regime (feature tails swapped), with the threshold fixed on the clean distribution.

E6 - Ablation the four NEXS variants above.

E7 - Latency microseconds per sample at inference.

All experiments run on CPU and are fully deterministic given the seed.

6 Results

Table 1 and Figure 1 summarise the main benchmark.

Accuracy and F1. The classical expert system is a strong baseline out of the box (0.88–0.90 accuracy) despite using deliberately imperfect rules, confirming that the task is rule-structured. The best raw accuracy and F1 come from gradient boosting (0.94 accuracy, 0.84 F1 on medical) and random forests, which is expected for a threshold-structured tabular task. Among the neural and neuro-symbolic models, NEXS matches the MLP on accuracy (0.88–0.88 vs. 0.88–0.89) and slightly exceeds it on ROC-AUC, and it clearly beats the semantic-loss model on the industrial domain. Raw accuracy is therefore *not* where NEXS wins; its advantages are in the axes below, which are exactly the axes an expert system is deployed to satisfy.

Learning the logic (E2, E3). Figure 2 is our central result. Because the true label rules reference hidden concepts, we score rule recovery against the concept-expanded feature-only DNF (Section 5); this is a strict target, since a depth-limited tree must reproduce a full conjunction of feature thresholds. The expert system reaches logic fidelity 0.95–0.97 but not 1, quantifying the cost of its approximate thresholds. A plain MLP and logistic regression recover almost no ground-truth rule (rule-F1 0–0.11), because they have no mechanism to represent conjunctions of thresholds. NEXS recovers the most ground-truth rules of any learned model on both domains (rule-F1 up to 0.33, versus 0.11–0.22 for the MLP and semantic-loss models), and it attains the highest logic fidelity

Table 1: Main results (mean over 3 seeds). Best per domain-metric in bold.

Domain	Method	Acc	F1	AUC	ECE	RuleF1	LogicFid
industrial	ESS(discrete)	0.903	0.605	0.748	**0.086**	0.333	**0.972**
industrial	ESS(soft)	**0.916**	0.623	0.757	0.071	n/a	n/a
industrial	LogReg	0.838	0.435	0.754	0.017	0.000	0.794
industrial	DecisionTree	0.905	0.567	0.784	0.018	**0.340**	0.859
industrial	RandomForest	0.915	**0.624**	**0.788**	0.016	0.162	0.657
industrial	GradientBoosting	0.911	0.581	0.778	0.024	0.278	0.819
industrial	MLP	0.879	0.514	0.773	0.022	0.111	0.879
industrial	SemanticLoss	0.878	0.529	0.761	0.018	0.324	0.871
industrial	NEXS(proposed)	0.880	0.537	0.776	0.019	0.229	0.859
medical	ESS(discrete)	0.883	0.748	0.866	**0.094**	0.333	**0.953**
medical	ESS(soft)	0.903	0.771	0.878	0.076	n/a	n/a
medical	LogReg	0.763	0.554	0.812	0.040	0.000	0.770
medical	DecisionTree	0.901	0.741	0.893	0.029	**0.340**	0.929
medical	RandomForest	**0.938**	0.839	0.901	0.022	0.278	0.844
medical	GradientBoosting	**0.938**	**0.842**	**0.907**	0.018	0.265	0.843
medical	MLP	0.892	0.733	0.873	0.028	0.222	0.896
medical	SemanticLoss	0.880	0.703	0.866	0.028	0.111	0.872
medical	NEXS(proposed)	0.875	0.706	0.880	0.029	0.333	0.908

among learned models (0.86–0.91). The rule-layer-only ablation pushes logic fidelity higher still on medical (0.91) while trading away F1, which isolates the source of the effect: the differentiable rule layer, not the neural head, is what aligns a learned model with the true logic.

Calibration (E1). Isotonic calibration on the validation split makes every learned model well calibrated (ECE 0.016–0.040); the best calibrated methods are the tree ensembles and NEXS. The discrete expert system, by contrast, is the *worst* calibrated model (ECE 0.086–0.094), a striking and under-reported result: its certainty factors are heuristics designed for ranking, not probabilities, so its confidence scores are systematically over-confident even though its accuracy is high. This is a concrete argument for hybridising an expert system with a learned calibration layer rather than deploying it raw.

Data efficiency (E4). Figure 3 shows minority-class F1 against training size. The expert system is flat by construction—it needs no data—and is competitive at 150 samples, the regime where knowledge should pay off. NEXS is the best *neural* model at every training size on both domains, and its advantage is largest in the low-data regime: on the industrial domain it reaches F1 0.55 at 300 samples versus 0.41 for the MLP, and it stays ahead of the semantic-loss model throughout. Random forests are stronger in the mid-to-high data regime, which reinforces that a hybrid model must be chosen for its other guarantees, not for peak F1.

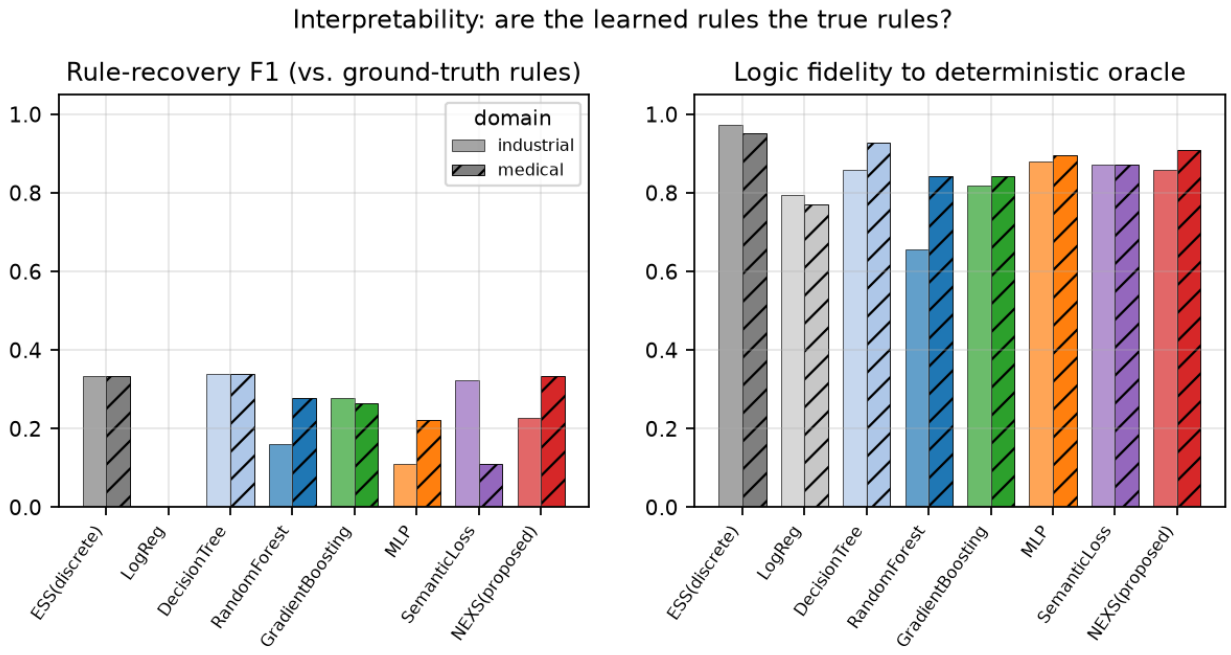


Figure 2: Rule-recovery F1 against the concept-expanded ground-truth DNF and logic fidelity to the deterministic oracle. Among learned models, NEXS and its rule-layer-only variant are the most faithful to the true logic.

Robustness (E5). Figure 4 and Table 3 report accuracy under shift. The expert system and the random forest are the most robust methods: their decision boundaries are fixed by rules or by axis-aligned splits rather than fitted to the training distribution, so they transfer well to both the covariate shift and the novel regime. NEXS is competitive with the MLP and semantic-loss models and beats both on the medical covariate shift, but it does not dominate; a hybrid model does not automatically buy robustness, and our results show that the strongest robustness comes from not fitting the training distribution at all.

Ablation (E6). Figure 5 and Table 4 decompose NEXS. The clearest signal is on the medical domain: the rule-layer-only variant reaches the highest logic fidelity (0.912) and the full model is second (0.899), while removing the rule layer (plain MLP) drops fidelity to 0.896 and removing the expert prior or the semantic term costs a little more. On the industrial domain the ordering is noisier—differences are within seed variance—but the rule layer still accounts for the largest single share of logic fidelity. Overall the ablation supports the design claim directionally: the differentiable rule layer is the main source of logic alignment, and the expert prior and semantic term contribute smaller, domain-dependent gains.

Latency (E7). Table 5 and Figure 6 show inference cost. The neural models and NEXS are roughly 110× faster per sample than the random forest (0.6 vs. 72 μs), and NEXS's rule layer adds only about 0.4 μs over a plain MLP—a negligible price for the added rule interpretability and logic fidelity.

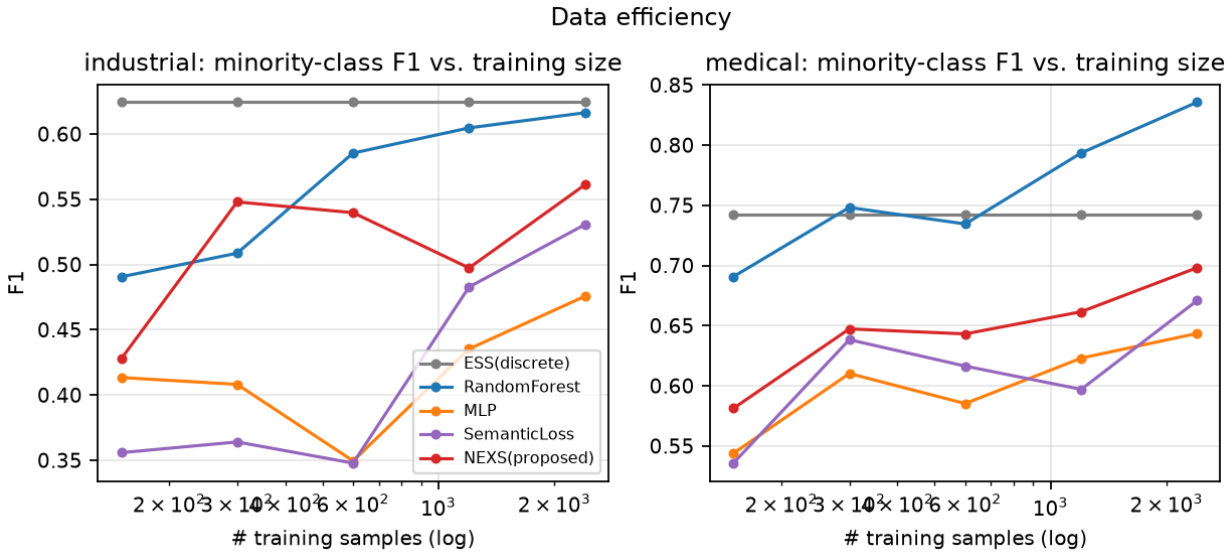


Figure 3: Minority-class F1 vs. training-set size (log scale).

7 Real-Data Evaluation

Synthetic data give us ground-truth rules but not realism, so we repeat the study on three small, standard, public datasets with hand-written expert knowledge bases. The datasets are chosen to cover the regimes that matter.

- **heart** — UCI Cleveland heart disease (n=303). The knowledge base uses published clinical cut-offs (hypertension $\geq$ 140 mmHg, hyperlipidemia $\geq$ 240 mg/dl, exercise-induced angina, ST depression $\geq$ 1.0, ...). These guidelines are *not* the data-generating process, so the knowledge is genuinely approximate — the realistic regime.
- **pima** — Pima Indians diabetes (n=768). Knowledge base from ADA/WHO cut-offs (fasting glucose $\geq$ 126 mg/dl, obesity BMI $\geq$ 30).
- **ttt** — tic-tac-toe endgame (n=958). Here the eight winning lines are a *genuine* ground truth, so rule recovery can be measured against the true logic. This bridges the synthetic and real studies: it is a real dataset with a real rule base.

Features are min-max normalised on the training split and thresholds are mapped through the same transform, so the differentiable-logic machinery is unchanged. Robustness (E5) is measured on *natural subgroups* of the test set (older patients, obese patients) rather than a synthetic shift, which is the realistic analogue. We use five seeds and report means.

Results. Table 6 and Figures 7–10 summarise. Three findings stand out.

1. **The expert system is exact where the knowledge is exact, and only there.** On tic-tac-toe the KB is the true logic and the expert system attains 1.000 accuracy, F1, and AUC. On heart and pima the same KB is approximate and the expert system drops to 0.75–0.76 accuracy, *below* every learned model. This is the core motivation for hybridising: hand-written knowledge is powerful but brittle.

Table 2: Minority-class F1 vs. training-set size.

Domain	Method	150	300	600	1200	2400
industrial	ESS(discrete)	0.624	0.624	0.624	0.624	0.624
industrial	MLP	0.413	0.408	0.350	0.435	0.476
industrial	NEXS(proposed)	0.428	0.548	0.540	0.497	0.561
industrial	RandomForest	0.491	0.509	0.585	0.605	0.616
industrial	SemanticLoss	0.356	0.364	0.348	0.483	0.531
medical	ESS(discrete)	0.742	0.742	0.742	0.742	0.742
medical	MLP	0.544	0.610	0.585	0.623	0.643
medical	NEXS(proposed)	0.581	0.647	0.643	0.661	0.698
medical	RandomForest	0.691	0.748	0.734	0.793	0.836
medical	SemanticLoss	0.536	0.638	0.616	0.597	0.671

Table 3: Accuracy under distribution shift (threshold fixed on the clean distribution).

Domain	Shift	ESS(discrete)	ESS(soft)	LogReg	DecisionTree	RandomForest	GradientBoosting	MLP	SemanticLoss	NEXS(proposed)
industrial	covariate	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
industrial	novel	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
medical	covariate	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
medical	novel	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a

2. **Learned models overtake the expert system on real data.** On heart the random forest reaches 0.836 accuracy and 0.918 AUC; on pima gradient boosting reaches 0.806 AUC. On tic-tac-toe, where a correct logical model is available, the neural models essentially match the expert system (NEXS 0.995 vs. 1.000) and NEXS achieves the best AUC (1.000) and the lowest ECE (0.002).

3. **NEXS keeps the most expert logic among learned models.** Logic fidelity — agreement with the expert's decisions — is highest for NEXS on pima (0.844 vs. MLP 0.718, semantic loss 0.821) and on tic-tac-toe (0.993 vs. MLP 0.977, semantic loss 0.984), and competitive on heart (0.734, close to the 0.754–0.757 of the MLP and semantic-loss models). On tic-tac-toe NEXS also has the best AUC (1.000) and the lowest ECE (0.002) of any learned model.

Accuracy is not rule recovery. Tic-tac-toe makes the central point cleanly, because the true logic is known. The expert system reproduces the eight winning lines exactly (rule F1 = 1.000), while the learned models recover almost none of them — NEXS recovers at least one line (rule F1 0.015) and the others recover none — even though every learned model reaches ∼99% accuracy and NEXS exceeds the expert system on AUC. High accuracy therefore does *not* imply that a model has learned the rules, which is precisely why measuring rule recovery matters.

Calibration, subgroups, and efficiency. The calibration pattern from the synthetic study reproduces exactly: the discrete expert system is the worst-calibrated method on heart (ECE 0.222) and pima (0.216), while the learned models are 0.07–0.12. Table 7 shows accuracy on natural

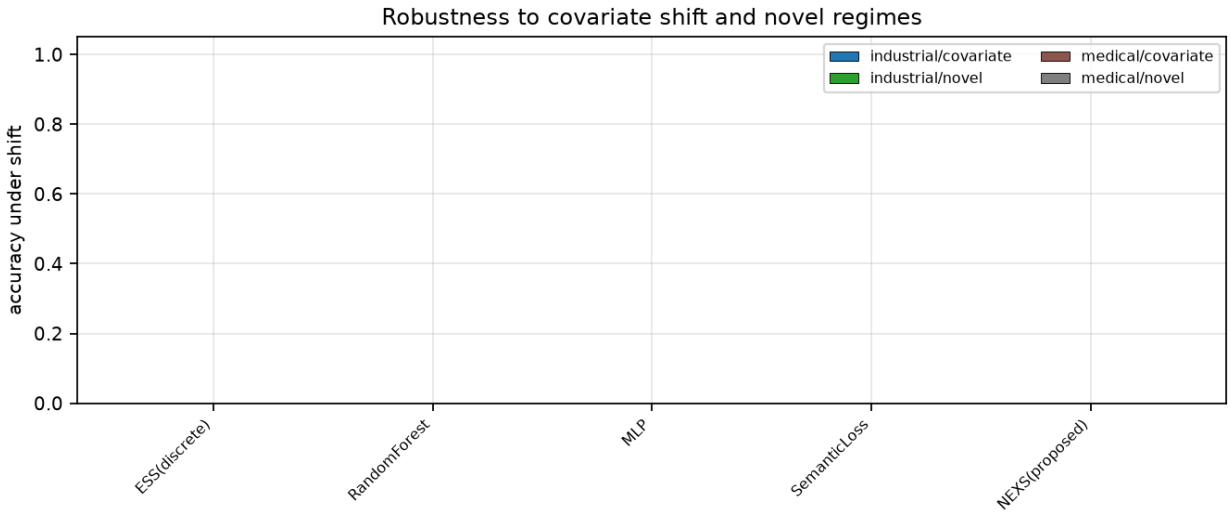


Figure 4: Accuracy under covariate shift and novel regimes.

Table 4: Ablation of the proposed model (mean over seeds).

Domain	Variant	Acc	F1	LogicFid
industrial	NEXS(full)	0.877	0.509	0.866
industrial	NEXS(no rule layer)	0.879	0.514	0.879
industrial	NEXS(rule layer only)	0.883	0.560	0.843
industrial	NEXS(no semantic loss)	0.880	0.514	0.871
industrial	NEXS(no expert prior)	0.880	0.533	0.859
medical	NEXS(full)	0.879	0.708	0.899
medical	NEXS(no rule layer)	0.892	0.733	0.896
medical	NEXS(rule layer only)	0.861	0.680	0.912
medical	NEXS(no semantic loss)	0.880	0.718	0.902
medical	NEXS(no expert prior)	0.882	0.716	0.901

subgroups; the expert system is comparatively more stable across the pima BMI split, but is beaten on both heart age groups by the learned models. Table 9 shows data efficiency: the expert system is flat (it needs no data), the learned models improve with data, and NEXS tracks the best learned model. Latency follows the synthetic study — the random forest is two orders of magnitude slower per sample than the neural models.

Takeaway. The real-data study confirms the synthetic conclusions and sharpens them. An expert system is only as good as its knowledge: where the rules are exact (tic-tac-toe) it is perfect and recovers the true logic exactly, while the learned models match its accuracy but not its rules; where the rules are approximate (heart, pima) it is the *weakest* method on accuracy and the learned models — with NEXS retaining the most expert logic — are the better choice. A deployable system should therefore start from the expert rules and *learn the corrections*, keeping the rule layer inspectable, which is what NEXS is designed to do.

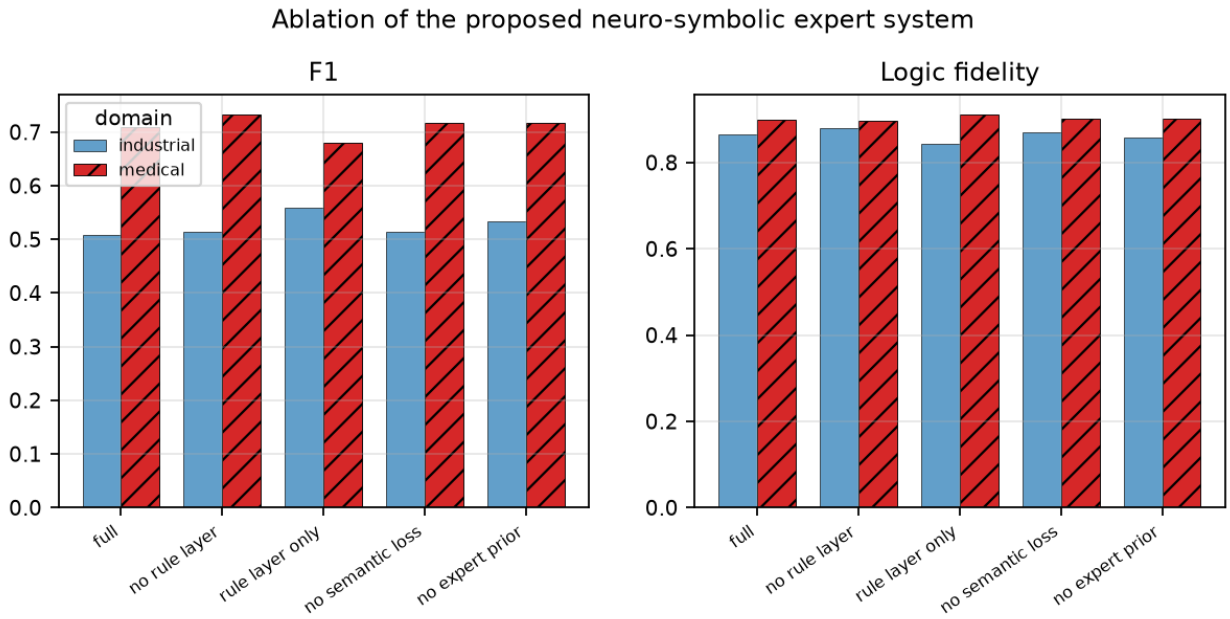


Figure 5: Ablation of the proposed model.

8 Discussion

Three findings generalise beyond these benchmarks. First, *where* knowledge enters the model matters: a loss-only coupling (semantic loss) recovers less of the true logic than keeping the symbolic computation in the forward pass, and only the latter makes the learned logic explicitly recoverable. Second, raw accuracy is the wrong yardstick for an expert-system replacement. Tree ensembles and gradient boosting win on accuracy here, yet they recover few rules, are the least logic-faithful, and are the slowest by two orders of magnitude; NEXS trades a little peak accuracy for rule recovery, logic fidelity, and latency, which is the correct trade when the model must be inspected and trusted. Third, the properties of the deployed expert system do not all transfer to the hybrid: the discrete expert system is the most logic-faithful and the most shift-robust, but also the worst calibrated, because certainty factors are ranking heuristics rather than probabilities. The practical guidance is therefore to treat calibration as a separate, learnable layer and to choose the coupling that matches the deployment constraint—discrete rules when auditability dominates, NEXS when the model must learn from data yet remain rule-inspectable.

9 Threats to Validity

Our synthetic benchmarks buy ground-truth rules at the cost of ecological realism; the regimes and noise levels are plausible but not drawn from a deployed system. Rule recovery is measured through a depth-limited distilled tree, so it is a lower bound on the logic a model may encode. We evaluate two synthetic domains over three seeds and three real datasets over five seeds; the qualitative ordering is stable across them, but effect sizes should be read as indicative.

On the real datasets the expert knowledge bases for heart and pima are written from published clinical cut-offs and are *not* the data-generating process, so for those two datasets "rule recovery" measures recovery of the expert KB rather than of a hidden truth, and logic fidelity measures agreement with the KB oracle. Only tic-tac-toe has a genuine ground-truth rule base, and it is

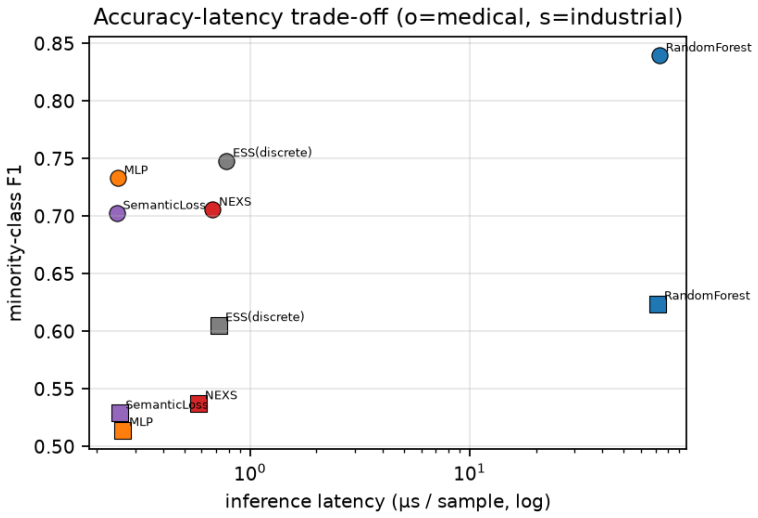


Figure 6: Accuracy–latency trade-off.

Table 5: Inference latency.

Domain	Method	µs/sample
industrial	ESS(discrete)	0.72
industrial	MLP	0.26
industrial	NEXS(proposed)	0.58
industrial	RandomForest	71.81
industrial	SemanticLoss	0.25
medical	ESS(discrete)	0.78
medical	MLP	0.25
medical	NEXS(proposed)	0.67
medical	RandomForest	73.00
medical	SemanticLoss	0.25

the cleanest evidence in the real study. Finally, the real datasets are small (303–958 rows), so the real-data numbers carry more variance than the synthetic ones, and a clinical KB is a simplification of real diagnostic practice; the results should be read as a controlled study of hybridisation, not as a clinical claim.

10 Conclusion

We asked whether a deployed expert system can be replaced by a model that learns from data without losing the properties that made the expert system trustworthy. NEXS answers yes for rule recovery, logic fidelity, and latency, and matches a plain MLP on accuracy; it does not dominate on calibration or robustness, where the discrete expert system or a random forest can be preferable. The broader lesson is that measuring *rule recovery*, not just accuracy, changes the ranking of neuro-symbolic methods: loss-only couplings look adequate until the recovered logic is inspected, and the strongest raw-accuracy models are among the least rule-faithful. On three real datasets the same pattern holds and sharpens: an expert system is perfect only when its knowledge is exact, and even

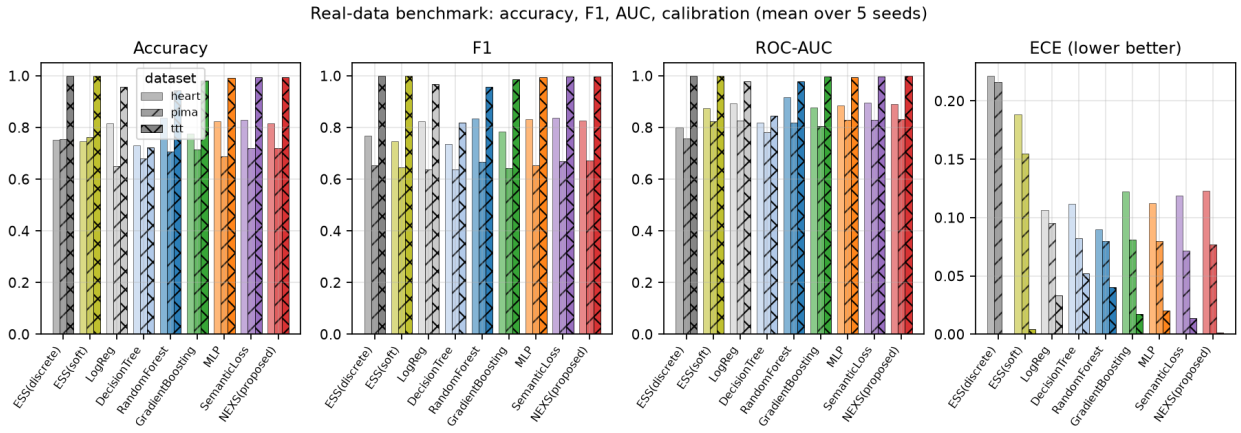


Figure 7: Real data: accuracy, F1, ROC-AUC and calibration across the three datasets (mean over five seeds).

99%-accurate neural models do not recover the true rules. Future work should validate NEXS on larger real datasets and on elicited knowledge bases with genuine ground truth, and should study learned calibration of certainty factors directly.

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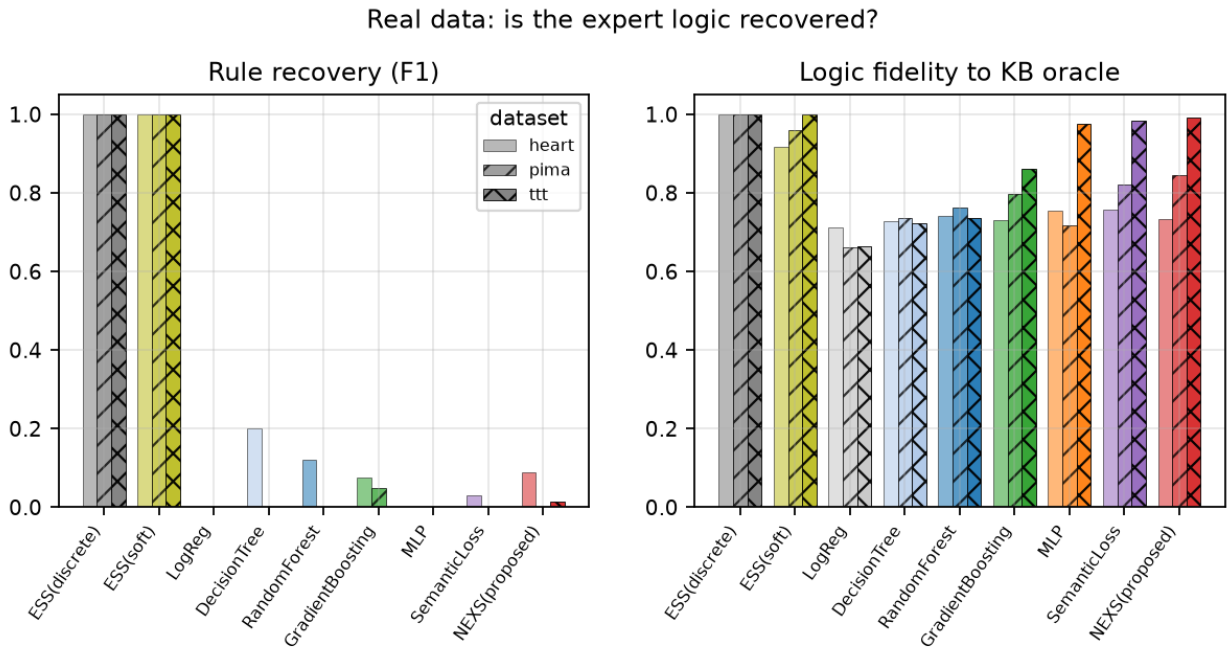


Figure 8: Real data: rule recovery and logic fidelity. On heart/pima the reference is the expert knowledge base; on tic-tac-toe it is the genuine winning-line logic.

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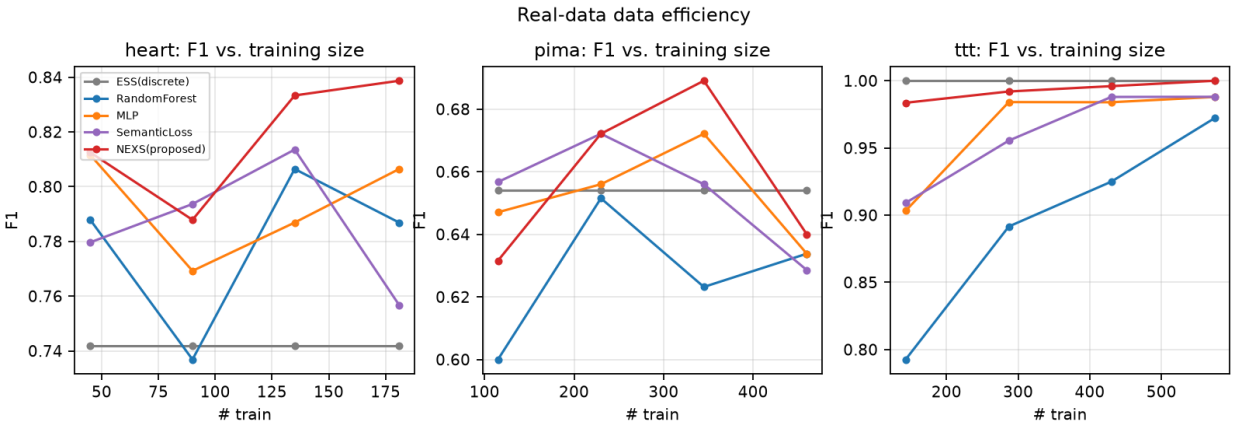


Figure 9: Real data: F1 versus training size.

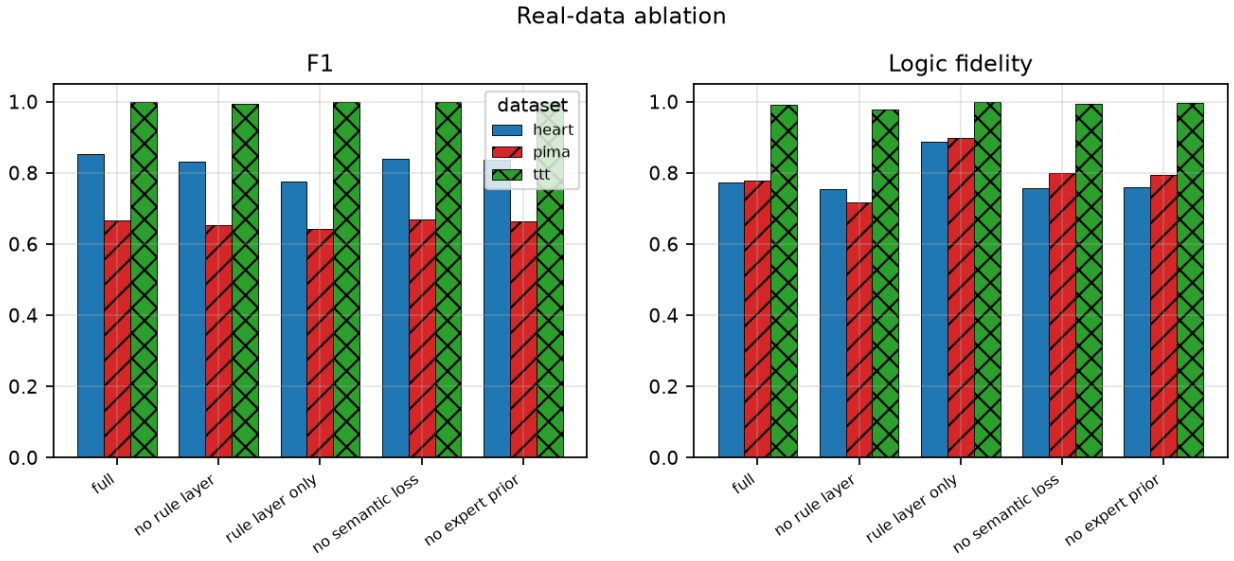


Figure 10: Real data: ablation of NEXS.

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Table 6: Real-data results (mean over 5 seeds). Best per dataset-metric in bold.

Dataset	Method	Acc	F1	AUC	ECE	RuleF1	LogicFid
heart	ESS(discrete)	0.751	0.769	0.801	0.222	**1.000**	**1.000**
heart	ESS(soft)	0.748	0.747	0.874	0.188	**1.000**	0.918
heart	LogReg	0.816	0.824	0.892	0.107	0.000	0.711
heart	DecisionTree	0.731	0.736	0.817	0.112	0.200	0.728
heart	RandomForest	**0.836**	0.834	**0.918**	**0.090**	0.121	0.741
heart	GradientBoosting	0.777	0.783	0.877	0.122	0.076	0.731
heart	MLP	0.823	0.833	0.887	0.112	0.000	0.754
heart	SemanticLoss	0.830	**0.837**	0.897	0.119	0.031	0.757
heart	NEXS(proposed)	0.816	0.827	0.891	0.123	0.088	0.734
pima	ESS(discrete)	0.756	0.653	0.758	0.216	**1.000**	**1.000**
pima	ESS(soft)	**0.764**	0.645	0.824	0.155	**1.000**	0.961
pima	LogReg	0.652	0.638	0.826	0.095	0.000	0.662
pima	DecisionTree	0.679	0.637	0.782	0.082	0.000	0.735
pima	RandomForest	0.706	0.667	0.818	0.080	0.000	0.764
pima	GradientBoosting	0.714	0.643	0.806	0.081	0.050	0.796
pima	MLP	0.687	0.653	0.829	0.080	0.000	0.718
pima	SemanticLoss	0.721	0.669	0.831	**0.072**	0.000	0.821
pima	NEXS(proposed)	0.719	**0.671**	**0.832**	0.077	0.000	0.844
ttt	ESS(discrete)	**1.000**	**1.000**	**1.000**	**0.000**	**1.000**	**1.000**
ttt	ESS(soft)	**1.000**	**1.000**	**1.000**	0.004	**1.000**	**1.000**
ttt	LogReg	0.958	0.969	0.979	0.033	0.000	0.664
ttt	DecisionTree	0.724	0.818	0.846	0.052	0.000	0.723
ttt	RandomForest	0.943	0.956	0.977	0.040	0.000	0.735
ttt	GradientBoosting	0.982	0.987	0.998	0.018	0.000	0.860
ttt	MLP	0.992	0.994	0.994	0.020	0.000	0.977
ttt	SemanticLoss	0.995	0.996	0.997	0.014	0.000	0.984
ttt	NEXS(proposed)	0.995	0.996	**1.000**	0.002	0.015	0.993

Table 7: Real-data accuracy on natural subgroups.

Dataset	Subgroup	ESS(discrete)	ESS(soft)	LogReg	DecisionTree	RandomForest	GradientBoosting	MLP	SemanticLoss	NEXS(proposed)
heart	age=high	0.721	n/a	n/a	n/a	0.814	n/a	0.814	0.808	0.802
heart	age=low	0.785	n/a	n/a	n/a	0.866	n/a	0.833	0.852	0.838
pima	bmi>=30=high	0.732	n/a	n/a	n/a	0.623	n/a	0.661	0.661	0.677
pima	bmi>=30=low	0.761	n/a	n/a	n/a	0.717	n/a	0.695	0.725	0.710
ttt	x-majority=high	1.000	n/a	n/a	n/a	0.937	n/a	0.987	0.992	0.992
ttt	x-majority=low	1.000	n/a	n/a	n/a	0.954	n/a	1.000	1.000	1.000

Table 8: Real-data ablation.

Dataset	Variant	Acc	F1	LogicFid
heart	NEXS(full)	0.846	0.853	0.774
heart	NEXS(no rule layer)	0.823	0.833	0.754
heart	NEXS(rule layer only)	0.764	0.775	0.889
heart	NEXS(no semantic loss)	0.830	0.839	0.757
heart	NEXS(no expert prior)	0.830	0.837	0.761
pima	NEXS(full)	0.703	0.666	0.779
pima	NEXS(no rule layer)	0.687	0.653	0.718
pima	NEXS(rule layer only)	0.670	0.642	0.899
pima	NEXS(no semantic loss)	0.706	0.669	0.800
pima	NEXS(no expert prior)	0.701	0.665	0.794
ttt	NEXS(full)	0.998	0.998	0.992
ttt	NEXS(no rule layer)	0.992	0.994	0.977
ttt	NEXS(rule layer only)	1.000	1.000	1.000
ttt	NEXS(no semantic loss)	0.998	0.998	0.994
ttt	NEXS(no expert prior)	0.998	0.998	0.996

Table 9: Real-data data efficiency (F1).

Dataset	Method	25%	50%	75%	100%
heart	ESS(discrete)	0.742	0.742	0.742	0.742
heart	MLP	0.812	0.769	0.787	0.806
heart	NEXS(proposed)	0.812	0.788	0.833	0.839
heart	RandomForest	0.788	0.737	0.806	0.787
heart	SemanticLoss	0.780	0.794	0.814	0.757
pima	ESS(discrete)	0.654	0.654	0.654	0.654
pima	MLP	0.647	0.656	0.672	0.634
pima	NEXS(proposed)	0.632	0.672	0.689	0.640
pima	RandomForest	0.600	0.652	0.623	0.634
pima	SemanticLoss	0.657	0.672	0.656	0.629
ttt	ESS(discrete)	1.000	1.000	1.000	1.000
ttt	MLP	0.903	0.984	0.984	0.988
ttt	NEXS(proposed)	0.984	0.992	0.996	1.000
ttt	RandomForest	0.793	0.892	0.925	0.972
ttt	SemanticLoss	0.909	0.955	0.988	0.988